

CD4⁺CD25⁺FOXP3⁺ regulatory T cells to protect against graft-versus-host disease

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Allogeneic hematopoietic cell transplantation (HCT) has benefited many patients with hematological malignancies and bone marrow failure states. However, graft-versus-host disease (GVHD) and immune incompetence remain significant challenges that limit the use of allogeneic HCT to patients with life-threatening, high-risk disorders because of the significant nonrelapse mortality associated with the treatment. With a greater understanding of the pathophysiology of GVHD, it has become apparent that this disorder represents a dysregulated immune reaction resulting in immune dysfunction, further exacerbated by treatment with

immunosuppressive medications and injury to lymphoid tissues where immune recovery and tolerance develop. The emergence of fundamental mechanisms of immune regulation, whose pioneers received the 2025 Nobel Prize in Physiology or Medicine, has resulted in new concepts for improving outcomes for patients and offers the hope that, with a reduction in transplant-related risk, HCT can be offered to more patients with a broader range of immune-mediated disorders. In this review, the history, preclinical studies, and successful translation of CD4⁺CD25⁺FoxP3⁺ regulatory T cells in the context of allogeneic HCT are discussed.

GVHD as a dysregulated immune reaction

Preclinical animal models have played a critical role in our understanding of transplant biology and graft-versus-host disease (GVHD) pathophysiology. Initially, canine and murine models, and, more recently, nonhuman primate models have been instrumental in establishing biological principles and guiding therapeutic interventions. It has long been recognized that donor-derived alloreactive T cells play a key role in GVHD pathophysiology.¹ To visualize these concepts, we developed the concept of bioluminescence imaging, in which a marker of cellular proliferation, in this case, the firefly luciferase (*luc*) gene, was inserted into donor T cells and then imaged noninvasively in recipient animals over time.² These studies demonstrated that within 24 to 48 hours, donor-derived T cells migrate to nodal sites, become activated, proliferate, and upregulate tissue-specific homing molecules required for entry into GVHD target tissues.³ Within several days, these activated T cells enter GVHD target tissues such as the skin, liver, and gastrointestinal tract (GIT), resulting in massive proliferation. These events do not elicit significant clinical symptoms until 3 to 4 weeks later, when the animals succumb to the effects of organ damage. These results highlighted that alloreactive T-cell activation occurs extremely early and is a time-dependent biological event.

Attempts to block GVHD in murine models were challenging and required the addition of monoclonal antibodies to block T-cell

entry into the GIT and to perform a splenectomy.⁴ These studies, although biologically informative, were not clinically practical. They highlighted that T-cell activation occurs at multiple priming sites and that activated alloreactive T cells were improperly regulated, resulting in ongoing proliferation, tissue infiltration, and organ damage. Within the conventional T-cell (Tcon) pool, several groups noted that naïve T cells were responsible for initiating GVHD, and depletion of this CD3⁺CD45RA⁺ cell population resulted in attenuated GVHD.^{5,6} This strategy has been translated into clinical practice with promising results.⁷

A central question was whether such events also occurred in humans. To address this question, we sequenced the T-cell receptor of T cells infiltrating the GIT in patients suspected of having GVHD. These studies demonstrated that several highly represented clones could be identified in the GIT and blood samples collected from patients with GVHD, which expanded in patients who failed treatment but retracted in those who responded to therapy.⁸ Interestingly, even though these patients were full donor chimeras, these T-cell clones were not found in the donor inoculum, suggesting that they were rare in the mobilized peripheral blood progenitor cells and underwent massive T-cell proliferation in vivo. A further validation of the importance of early events in GVHD initiation comes from the timing of posttransplant cyclophosphamide administration, which is given on days 3 and 4, time points at which alloreactive T cells are undergoing significant proliferation and are likely susceptible to cell cycle-active chemotherapy.⁹

Immune regulatory cells

Control of immune reactions and maintenance of immune homeostasis are regulated by a complex array of interrelated mechanisms. Central to the control of immune tolerance are populations of immune regulatory cells, of which the most important is likely the CD4⁺CD25⁺FoxP3⁺ regulatory T cells (Treg).^{10,11} These cells were initially characterized by Sakaguchi, who demonstrated that Treg potently suppressed the mixed lymphocyte reaction (MLR) *in vitro*, a hallmark of Treg function.¹² These discoveries resulted in the 2025 Nobel Prize in Physiology or Medicine being awarded to Mary Brunkow, Fred Ramsdell, and Shimon Sakaguchi. Patients with mutations in FoxP3 have immune dysregulation, polyendocrinopathy, enteropathy, X-linked syndrome, and mice have the *scurfy* phenotype, further documenting the importance of this population of cells.¹² Although canonical, CD4⁺CD25⁺FOXP3⁺ Treg are not the only population of T cells capable of general immunoregulation or of modulating and preventing GVHD. Other populations include invariant natural killer T cells,^{13,14} Type 1 regulatory cells,¹⁵ and CD8 regulatory cells. Other non-T-cell populations are also implicated in playing important roles, including B regulatory cells, myeloid-derived suppressor cells, dendritic cell populations among others. Given the importance of immune regulatory mechanisms, it is not surprising that there are so many overlapping populations of cells responsible for the control of aberrant immune reactions and the restoration of immune homeostasis. Further, there is evidence for cross talk among these different immune regulatory cell populations forming cellular networks that help to reinforce each other, and the establishment of these networks is likely a key condition for healthy vs unhealthy immune reconstitution following hematopoietic stem cell transplantation (HSCT)¹⁶ or in other settings where sustained immune tolerance is needed.¹⁷

The role of Treg in murine GVHD has been extensively evaluated in both acute and chronic manifestations. Initially, studies that focused on the depletion of Treg demonstrated exacerbation of GVHD and more rapid animal lethality.¹⁸ Subsequently, a number of groups focused on the adoptive transfer of Treg in different strain combinations across both major and minor histocompatibility barriers, in which Treg consistently suppressed GVHD pathophysiology and lethality.¹⁹⁻²² Our group and others demonstrated that after the adoptive transfer of Treg and Tcon, the beneficial graft-versus-tumor (GVT) effects were maintained.^{19,23}

The impact of Treg in these model systems was investigated in important ways that informed subsequent clinical translation. First, it was demonstrated that adjusting the Treg:Tcon ratio to 1:1 with respect to cell numbers was optimal for suppression of GVHD with maintenance of GVT effects. Second, administering Treg 48 hours before Tcon allowed for a reduction in the Treg dose and a more potent biological effect, because Treg could expand *in vivo* before the administration of Tcon.²⁴ Interestingly, administering Treg 48 hours after Tcon rendered them ineffective. Third, the administration of Treg resulted in less damage to lymphoid organs such as lymph nodes, thymus, and spleen, resulting in more effective immune recovery from donor-derived T cells in Treg-treated animals developing from HSCs. In these studies, Treg-treated animals had accelerated

B- and T-cell recovery and were more capable of mounting an effective immune response to an infectious agent such as cytomegalovirus.^{23,25}

Mechanistically, it remains unclear exactly how Treg affect these important biological reactions. Treg need to migrate to lymphoid tissues to exert a biological effect, because ~50% of murine Treg express CD62L, allowing for lymphoid homing. Although both populations effectively suppress the MLR, only the CD62L⁺ Treg were active *in vivo*.^{26,27} Treg potently suppress Tcon proliferation *in vivo*. Through sequencing of the T-cell receptor and RNA-sequencing analysis, it was determined that Treg primarily impact gene expression in CD4⁺ T cells, particularly genes involved in T-cell activation and oxidative metabolism.²⁸ It is thought that the suppression of Tcon proliferation is central to the control of GVHD, whereas GVT effects do not require the same degree of Tcon proliferation. Further, the trafficking patterns of Tcon in the presence of Treg may be affected, thereby augmenting their ability to reject tumors, for example, in leukemia, in which the disease is localized to the bone marrow.²⁹ Correlative studies support these concepts, showing that donors who have higher numbers of Treg in the PBPCs grafts have a lower risk of GVHD and improved survival.³⁰

Clinical translation

These provocative preclinical findings that the timed addition of purified Treg and Tcon resulted in the control of GVHD, retention of graft-versus-leukemia (GVL) effects, and improved immune reconstitution led a number of groups to develop strategies for translation. The group at the University of Minnesota focused on *ex vivo* expansion of Treg from umbilical cord blood (UCB) to augment the number of Treg delivered during transplantation. In their studies, UCB Treg were expanded from a dedicated cord by purification of Treg and then either by expansion with CD3×CD28 beads followed by interleukin-2 (IL-2) alone or with engineered irradiated modified K562 cell lines.³¹ After 14 to 21 days in culture, Treg expanded significantly. Another strategy used by groups at the University of Perugia and Stanford University was to purify Treg from either unmobilized or granulocyte colony-stimulating factor-mobilized peripheral blood progenitor cells after isolation of CD34⁺ stem cells by immunomagnetic selection. The Treg were purified from the flow-through by CD25⁺ immunomagnetic selection.³² The Stanford University group purified Treg by high-speed cell sorting using the observation that Treg express CD4 and CD25, but have low expression of the IL-7 receptor CD127.^{33,34} These CD4⁺CD25⁺CD127^{low} cells were purified to 95% to 99% purity and uniformly expressed FoxP3. The cells were potent suppressors of the MLR. Initial clinical trial evaluation with cryopreserved and then thawed purified Treg demonstrated a lack of GVHD prevention when compared with early results for fresh Treg. In a bedside-to-bench evaluation, it was found that the freeze thaw of Treg reduced their *in vivo* function through a metalloprotease that degraded CD62L, which is required for appropriate lymph node homing, with the cleavage of other immune receptors also probably contributing to additional functional loss.²⁸ The immunomagnetic-selected or highly purified fresh Treg were used for clinical translation by the groups at Perugia and Stanford, respectively. An in-depth characterization of these

and other Treg clinical trials to prevent and treat GVHD has recently been published.³⁵

Clinical experience with expanded cord blood- and peripheral blood-induced Treg

The University of Minnesota completed a 23-patient, open-label, dose-escalation phase 1 study of ex vivo-expanded Treg in individuals undergoing 4-6/6 HLA-matched UCB transplantation ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT00602693) identifier: NCT00602693). Treg were expanded from a third-party 4-6/6 HLA-matched cord blood unit and were administered fresh on day +1 after UCB transplantation with reduced-intensity conditioning (RIC), with some patients receiving additional cryopreserved Treg on day +15. The study showed manufacturing feasibility and safety up to the maximum dose of Treg, with grade II to IV GVHD rates of 43% and a chronic GVHD rate of 14%, which was reduced compared with that of a historical control cohort.³⁶ Although graft loss was observed in 2 patients, there appeared to be no exceptional infectious disease risk, and the cumulative relapse incidence was 32%. A later phase 1 study in 11 patients with a modified method for Treg expansion with CD64/CD86-expressing K562 feeder cells showed similar results with respect to GVHD incidence and tolerability.²⁶ A parallel 16-patient phase 1 trial study in myeloablative HLA-matched transplantation explored induced FOXP3⁺ Treg that were culture expanded from CD4⁺CD25⁻ fraction of a peripheral blood apheresis product³⁰ ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT01634217) identifier: NCT01634217). The incidence of GVHD did not appear substantially different from that observed in historical controls, and the trial was stopped because of concerns regarding feasibility of manufacturing. Treg cells derived from the human thymus or induced pluripotent stem cells may also be a source of Treg for clinical translation and GVHD prevention.^{37,38}

Immunomagnetic bead-enriched primary Treg from donor peripheral blood apheresis

From the 1980s to 2011, first-generation efforts to reduce GVHD through donor graft selection were almost entirely focused on controlling T-cell numbers as a single parameter. This was, in large part, due to the technical limitations of anything more advanced than positively selecting CD34⁺ HSCs or negatively selecting T cells by magnetic bead selection of a single marker. The exploration of first-generation graft engineering approaches demonstrated that very low numbers of T cells were needed to reduce GVHD risk ($<1 \times 10^5$ T cells per kg recipient body weight) and, even then, for haploidentical transplant GVHD rates remained substantial.

In 2011, the Perugia group launched the early evaluation of “second-generation grafts” by successfully selecting donor Treg using CD25⁺ magnetic beads and administering them 4 days before CD34⁺ HSCs and Tcon in a 28-patient phase 1 clinical trial of haploidentical recipients with high-risk leukemia.³² No posttransplant immunosuppression was administered. The incidence of grade II to IV acute GVHD was 8%, with

no chronic GVHD and low relapse rates observed. A 43-patient follow-up phase 2 trial by the same group showed a grade II to IV acute GVHD incidence of 15% and a chronic GVHD incidence of <2%, with exceptional relapse-free survival (RFS), suggesting that this strategy preserved the GVL effect.³⁹ Some patients received T-cell-depleting antithymocyte globulin or alemtuzumab therapy as part of conditioning. In both studies, substantial nonrelapse mortality (NRM) at 1 year, mainly due to infections, was observed, which was commensurate with standard institutional experiences at the time. More recently, the group reported the results of a phase 2 study with an innovative radiation conditioning approach using age-defined dosing of total body irradiation or total marrow/lymphoid irradiation in 50 patients with acute myeloid leukemia.⁴⁰ No T-cell-depleting agents or posttransplant immunosuppression was given. The incidence of acute grade II to IV GVHD was 30%, and there was a 2% incidence of moderate-to-severe chronic GVHD, with NRM of 21% and very low relapse.

This pioneering translational effort in haploidentical transplantation to use Treg to prevent GVHD has resulted in 2 generally important observations. The first is that Treg appear to facilitate immune reconstitution that favors GVL immune states in bone marrow and lymphoid tissues, which may help account for the remarkably low relapse rates observed.^{34,35} The second is that when acute GVHD occurs after both HLA-matched and haploidentical Treg donor grafts, it is generally lower grade and more responsive to primary steroid therapy than GVHD manifested after conventional immunosuppression used for GVHD prophylaxis. The reasons for this are unclear, although 1 can hypothesize that in the presence of Treg the massive expansion of Tcon is attenuated and strategies to decrease T-cell number and function with, for example, corticosteroids may be more effective. However, further studies are required to confirm and understand this observation.

Highly purified second-generation donor Treg products for GVHD prevention

A potential limitation of immunomagnetic bead selection of CD25⁺ cells to enrich for Treg is that the actual quantity of CD4⁺CD25⁺CD127^{low} Treg being administered can be highly variable with bead enrichment alone. In preclinical studies, the ratio of Treg:Tcon was important for GVHD prevention, so a manufacturing process for producing highly purified Treg by clinical-grade flow cytometry was pursued. In addition to improved precision and likely more reproducible immune reconstitution, the isolation of high-purity Treg also facilitates the future development of the platform through cell expansion or genetic modification.

In 2019, the Stanford group launched the evaluation of “second-generation high-precision grafts” with the report of a 19-patient, open-label, phase 1 dose-escalation study in 10/10 or 9/10 HLA-matched recipients undergoing myeloablative conditioning for leukemia, myelodysplastic syndrome, or high-risk lymphoma³⁴ ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT01660607) identifier: NCT01660607). After the initial patient experience, this phase 1 study underwent a protocol modification that changed from an initial strategy of cryopreserved Treg with no GVHD prophylaxis to

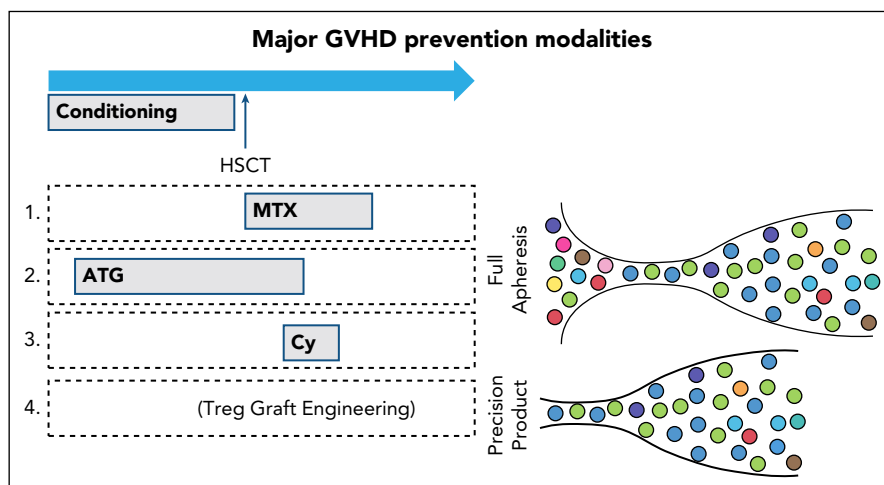


Figure 1. Conventional vs Orca-T transplant grafts. Comparison of traditional GVHD prevention strategies for mobilized unmodified HSCs and progenitor cells and Orca-T consisting of CD34⁺ purified HSCs and progenitor cells, Treg, and Tcon. ATG, antithymocyte globulin; Cy, cyclophosphamide; MTX, methotrexate.

the use of fresh Treg with single-agent immunosuppression.⁴¹ This phase 1 study resulted in the dose that has been used in subsequent studies, consisting of CD34⁺ cells ($>2 \times 10^6/\text{kg}$) and Treg (2×10^6 to $3 \times 10^6/\text{kg}$) on day 0 followed by donor Tcon ($3 \times 10^6/\text{kg}$) on day +2 to 3 after hematopoietic cell transplantation (Figure 1). In patients receiving this dose, very low levels of acute and chronic GVHD were observed. It should be noted that the dose of Treg of 2×10^6 to $3 \times 10^6/\text{kg}$ is based more upon feasibility compared with toxicity, given that this is the dose that can be reliably purified without ex vivo expansion.

Based on the phase 1 results, an open-label, 2-stage, phase 2 single-center study was initiated.⁴² In the first stage, 24 patients were randomized to 2 arms to evaluate acute GVHD incidence with single-agent tacrolimus (Tac) or sirolimus prophylaxis vs no prophylaxis. The no prophylaxis group failed to show noninferiority to the single-agent prophylaxis group, which had a grade II to IV acute GVHD incidence of 58% vs 8%, respectively. As an important note, all patients with no GVHD prophylaxis who developed acute GVHD responded to primary steroid therapy, further supporting the concept that when GVHD occurs, it is more responsive to therapy. In this cohort of patients, there was no relapse of their high-risk primary disease.

Based on the stage 1 results of the phase 2 trial, the second-stage study evaluating efficacy in 44 patients was completed.⁴² This showed a grade II to IV acute GVHD incidence of 25%, with 91% patients responsive to primary steroid therapy alone and an incidence of moderate-to-severe chronic GVHD of 11%. The trial met its end point of improved GVHD-free relapse survival (GRFS) of 64%, with a notably low 1-year NRM of 4.5%.

In parallel with the phase 2 single-center trial, the Stanford group Treg approach was commercialized and tested in an open-label, multicenter, phase 1b trial of Orca-T (ClinicalTrials.gov identifier: NCT04013685). This trial showed that the results achieved at a single center were broadly scalable and that centralized manufacturing for related and unrelated donor

apheresis products was feasible and safe. Interim reports indicate a GRFS of 72%, RFS of 78%, and overall survival of 91% at 1 year.

In 2021, a phase 3 trial comparing Orca-T with the standard-of-care regimen was initiated, the results of which have been recently reported in *Blood*⁴³ (ClinicalTrials.gov identifier: NCT0531670). This was a multicenter, 2-arm, randomized, controlled trial conducted at 19 participating HSCT centers. The trial compared Orca-T with single-agent Tac ($n = 93$) prophylaxis vs an unmanipulated donor apheresis product with Tac and methotrexate (Tac/MTX; $n = 94$). The Tac/MTX comparator arm was chosen because this was the most common GVHD prophylaxis strategy reported to the Center for International Blood and Marrow Transplant Research at the time of study initiation and was the approach supported by the US Food and Drug Administration. The trial met its primary end point early, demonstrating superior chronic GVHD-free survival at 1 year of 78% for Orca-T and 38% for Tac/MTX, with chronic GVHD defined as moderate to severe by the 2014 National Institutes of Health consensus criteria.⁴⁴ A number of secondary end points at 1 year favored Orca-T vs Tac/MTX, including GRFS (63% vs 31%) and NRM (3.4% vs 13%), with a trend toward improved overall survival (94% vs 83%). The myeloablative conditioning regimen consisted of busulfan, fludarabine, and thiopeta and is considered a generally intensive approach, which likely accounts for the overall high RFS across both arms (72% vs 73%). Measurement of immune function between the arms showed a likely more reproducible T-cell reconstitution in the Orca-T arm, as well as a statistically increased proportion of Treg in the peripheral blood early in immune reconstitution. T-cell numbers in the peripheral blood of patients receiving Tac/MTX were substantially higher early in immune reconstitution, probably reflecting a “saturation” of T cells from the unmanipulated donor product. Because of this and/or possibly due to the greater amount of immunosuppression, the Tac/MTX group showed a substantially higher incidence of grade ≥ 3 infections of 16.1% vs 8.4% for Orca-T.⁴³ Of note, Tac/MTX prophylaxis had previously had the lowest incidence of infections compared with other GVHD prevention modalities, such as posttransplant cyclophosphamide and the

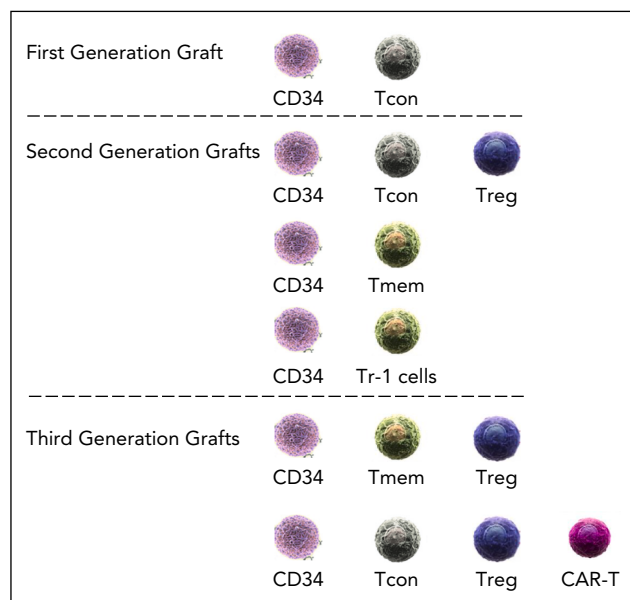


Figure 2. Generations of graft engineering. Different graft engineering approaches with progressively more sophisticated cell population selection strategies: first-generation grafts involving selection of CD34⁺ HSCs and bulk T cells (Tcon); second-generation grafts involving selection of CD34⁺ cells, Tcon, and Treg vs CD34⁺ cells, Tmem vs CD34⁺ cells, and Tr-1 regulatory cells; and third-generation grafts with more sophisticated cell selection strategies. Tmem, memory T cells; Tr-1, type 1 regulatory cells.

use of first-generation CD34-selected grafts.⁴⁵ A recent retrospective single-center study also suggests substantially different degradation of renal function by conditioning GVHD prophylaxis support, with Orca-T showing the least functional loss.⁴⁶

Because the peak incidence of many transplantable hematological malignancies occurs later in life, additional studies of this concept in the setting of RIC or nonmyeloablative conditioning are important. We have just concluded enrollment in a single-center, open-label, phase 1 study of Orca-T in 8/8 and 7/8 HLA-matched transplantation with RIC ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT01203722) identifier: NCT01203722), in which conditioning intensity was reduced over time in 3 cohorts, and initial results have been presented (Villar-Prados). All patients have shown engraftment, with full donor myeloid chimerism achieved in 100% of patients, and a low incidence of both acute and chronic GVHD. The least intense RIC, consisting of fludarabine (40 mg/kg), thiopeta (5 mg/kg), and 300 cG total body irradiation has been successfully transitioned to the outpatient setting. An open-label, phase 2 multicenter testing approach is now open and accruing.

Third-generation donor Treg products for GVHD prevention

The success of Orca-T as a second-generation graft is promising and warrants further extensive evaluation and yet, even as Orca-T potentially achieves commercialization, the field is already evaluating so-called “third-generation” allogeneic donor products in which donor cell composition is defined along ≥3 dose

parameters (Figure 2). An example is an investigator-led study at Stanford University in which investigators have successfully integrated Orca-T followed by allogeneic anti-CD20/22 chimeric antigen receptor (CAR) T cells for the treatment of high-risk relapsed/refractory B-cell acute lymphoblastic leukemia, with early results showing remarkable safety and tolerability, as well no relapse thus far, and all patients achieving minimal residual disease negativity, with the caveat that a proportion of patients have lasting B-cell aplasia after transplantation ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT05507827) identifier: NCT05507827).

As another example of a third-generation allogeneic donor product, Orca-Q product consists of CD34⁺ selected HSCs, Treg, and also T-cell subsets that exclude naïve T cells. Unlike Orca-T, Orca-Q is administered as a single product. Orca-Q is being evaluated in 6-7/8 HLA-matched and haploidentical transplantation. Early results suggest a low incidence of acute and chronic GVHD in HLA-matched patients without any other GVHD prophylaxis. However, the most studied population is haploidentical patients, and results from an ongoing multicenter phase 1 trial suggest a low incidence of acute and chronic GVHD with a GFRS of 82% for patients with myeloid malignancies or acute lymphoblastic leukemia ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/study/NCT03802695) identifier: NCT03802695; Srour et al). The phase 1 data suggest a remarkably low NRM of 2% to 3% across all donor matches and, for this reason, a multicenter phase 1 trial testing Orca-Q in HLA-matched recipients for the treatment of multiple sclerosis is underway, which is of particular interest because of its potential for replacement of recipient microglial cells with donor cells, thereby offering immune modulation beyond other cell therapy modalities such as CAR T-cell therapy currently in clinical trial evaluation.

In vivo activation and expansion of Treg

HSCT is exceptionally well suited for the implementation of adoptive transfer of Treg because a donor graft is required as part of the process. Yet, the in vivo expansion and modulation of Treg are also viable strategies with active research, both in combination with adoptive transfer cell therapy or on their own. There are a number of different approaches to achieve this, and the most notably advanced approach currently in clinical use is the administration of recombinant human IL-2.^{47,48} Other approaches include protein-engineered cytokines,⁴⁹ agonism of the TNFR2 receptor,⁵⁰ blockade of TIM-1,⁵¹ and JAK inhibition.⁵²

In addition to the expansion of Treg, how Treg interact with and generate other regulatory cell populations is likely important for improved immune reconstitution that prevents GVHD. The early evaluation of immune reconstitution after administration of second-generation Orca-T shows an increased distinct appearance of CD4 expressing Tcon, that do not express CD25 but do express FOXP3, and have immunoregulatory properties.⁵³ The further development of surrogate immune biomarkers of healthy immune reconstitution could allow for increased effectiveness in preventing GVHD and also potentially minimize the need for immunosuppressive agents or even promote GVL effects.

Future directions

The use of Treg cell therapy to prevent acute and chronic GVHD and facilitate balanced immune reconstitution after HSCT stands as the most mature application of Treg therapy and forms the basis of the first commercial Treg cell therapy product potentially to be approved this year. Given the extraordinary tolerability and exceptionally low toxicity and NRM, there is excitement to test this approach in nonmalignant indications, such as the establishment of immune tolerance in combined organ and HSCT or the treatment of autoimmune diseases. Likewise, we anticipate that other CAR T-cell therapies and approaches to enhance GVT effects will begin testing in other diseases, such as mantle cell lymphoma or multiple myeloma.

Because Treg cells are a “living drug,” there is also excitement to further define Treg subsets that are more stable or effective. Further, many exciting preclinical strategies to genetically modify Treg are being explored that could potentially greatly enhance their performance in HSCT, including ways to selectively expand Treg, introduce an orthogonal IL-2 receptor that allows for selective expansion with an orthogonal IL-2 protein “immunoshield” Treg from antibody depletion,⁴⁹ engineer chemokine receptors for targeted trafficking,⁵⁴ and generate CAR Treg.^{55,56} These and other strategies hold promise to bring these immunological concepts forward to the clinic to treat a broad range of immune disorders characterized by dysregulated immune responses, including cancer, autoimmunity, and transplant rejection. With these advances, we look forward to improving results of allogeneic transplantation with reduced toxicity and NRM for the treatment of hematological malignancies and to expansion of transplantation approaches

to other disorders that have been limited by treatment-related toxicity.

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Authorship

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Footnote

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